

Q

$$y = \beta_0 + \beta_1 x$$

$$\log(y) = \beta_0 + \beta_1 \log(x)$$

After Regression Analysis, we have

$$\log(y) = 1.5 + 3.3 \log(x)$$

In order to interpret the coefficient - 3.3, we need to take derivative on both sides

$$\frac{d[\log(y)]}{d\log(x)} = 0 + 3.3 \frac{d[\log(x)]}{d\log(x)}$$

$$\frac{d \log(y)}{d \log(x)} = 3.3$$

Solving left side of equation using chain rule of derivatives,

$$\left(\frac{dy}{y}\right) \left(\frac{x}{dx}\right) = 3.3$$

$$\frac{dy}{y} = \frac{dx}{x} (3.3)$$

In terms of % age

$$100 \cdot \frac{dy}{y} = 100 \cdot \frac{dx}{x} (3.3)$$

$$\% \text{ change in } y = \% \text{ change in } x (3.3)$$

It means if x is increased by 1%, y will increase by 3.3%.